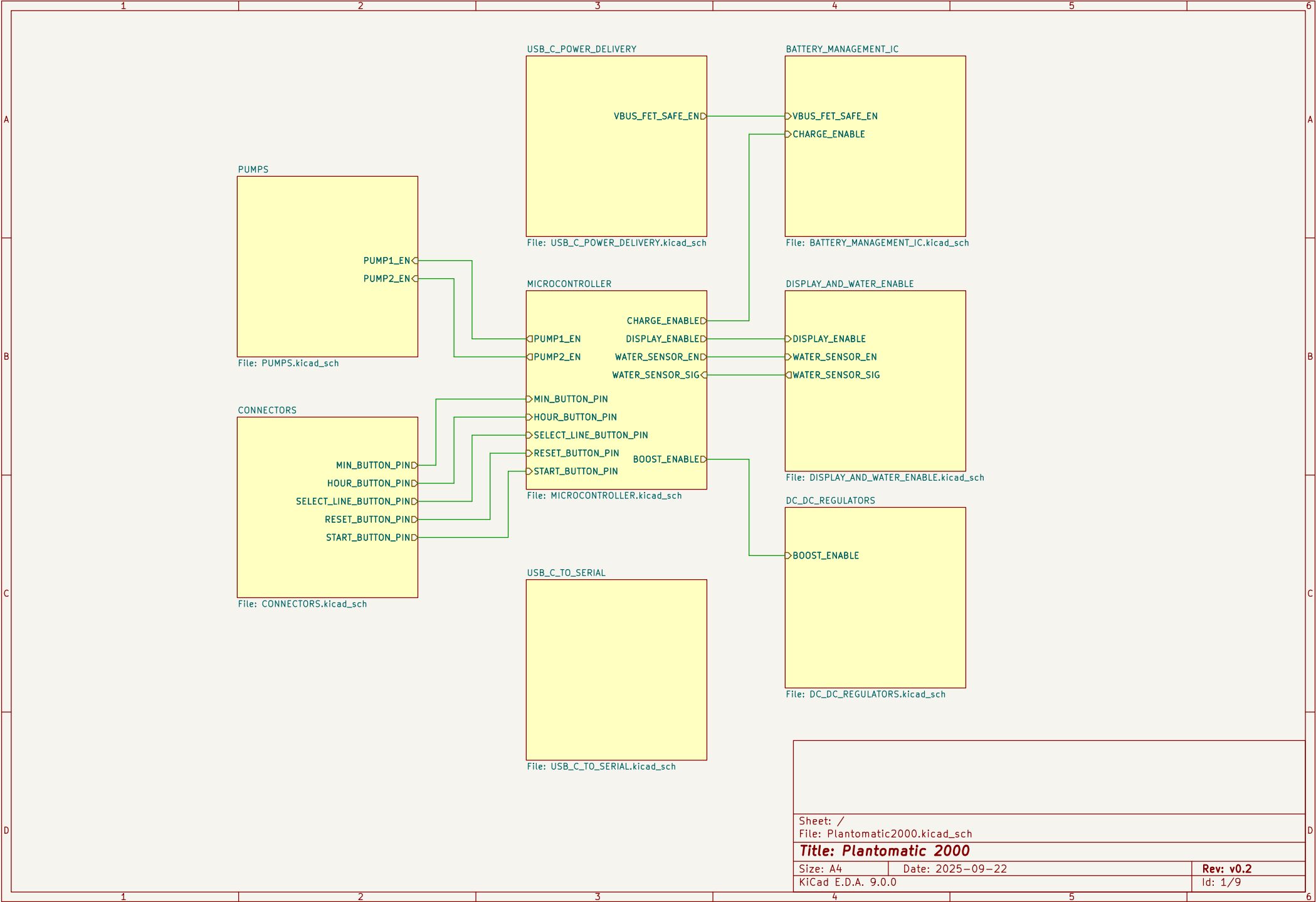
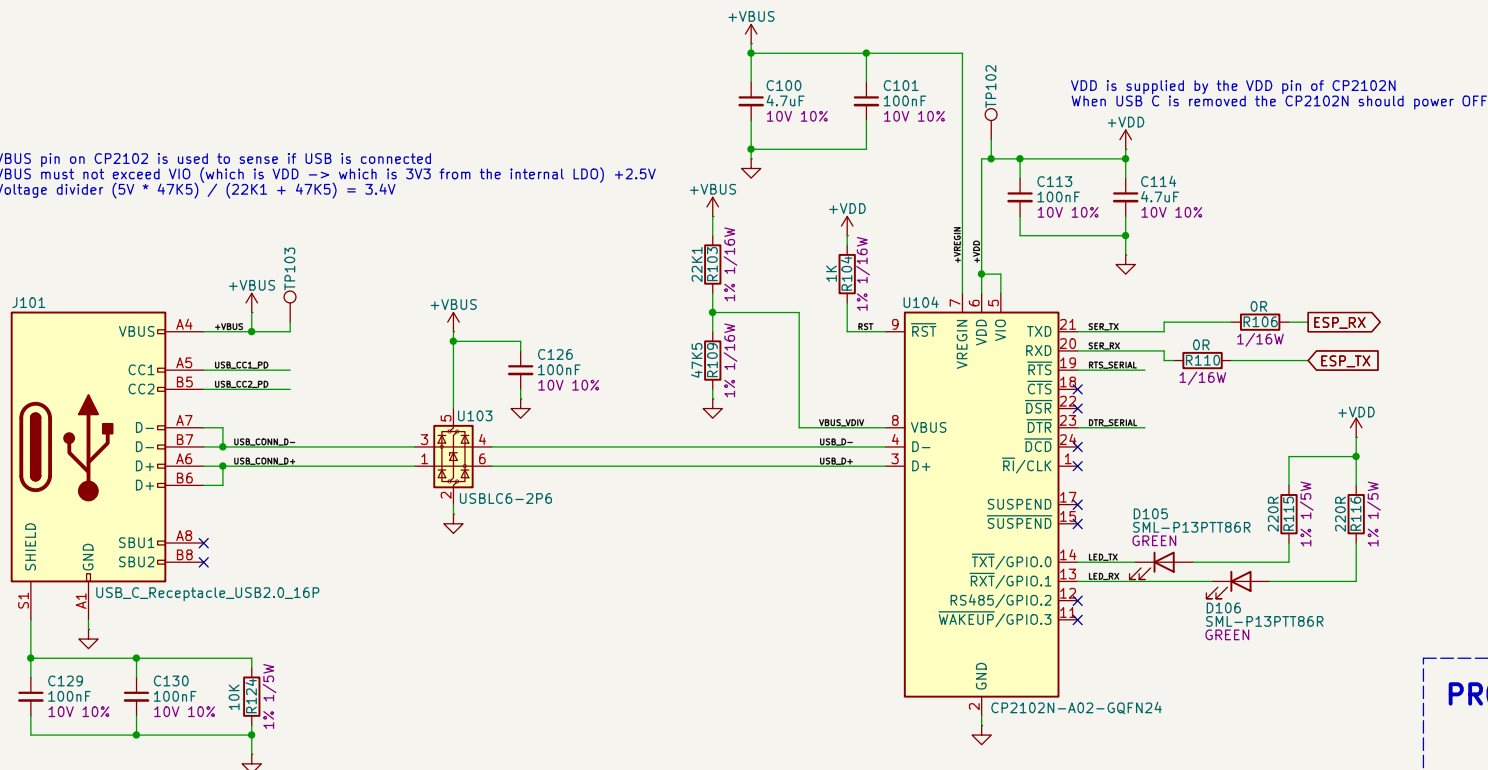


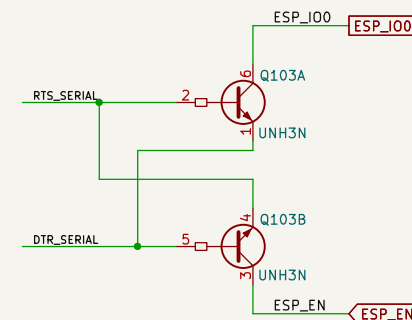


CP2102N USB TO SERIAL CONVERTER

VBUS pin on CP2102 is used to sense if USB is connected
VBUS must not exceed VIO (which is VDD -> which is 3V3 from the internal LDO) +2.5V
Voltage divider $(5V \cdot 47K5) / (22K1 + 47K5) = 3.4V$



PROGRAMMING BOOT & RESET



Sheet: /USB_C_TO_SERIAL/
File: USB_C_TO_SERIAL.kicad_sch

Title: USB_C_TO_SERIAL

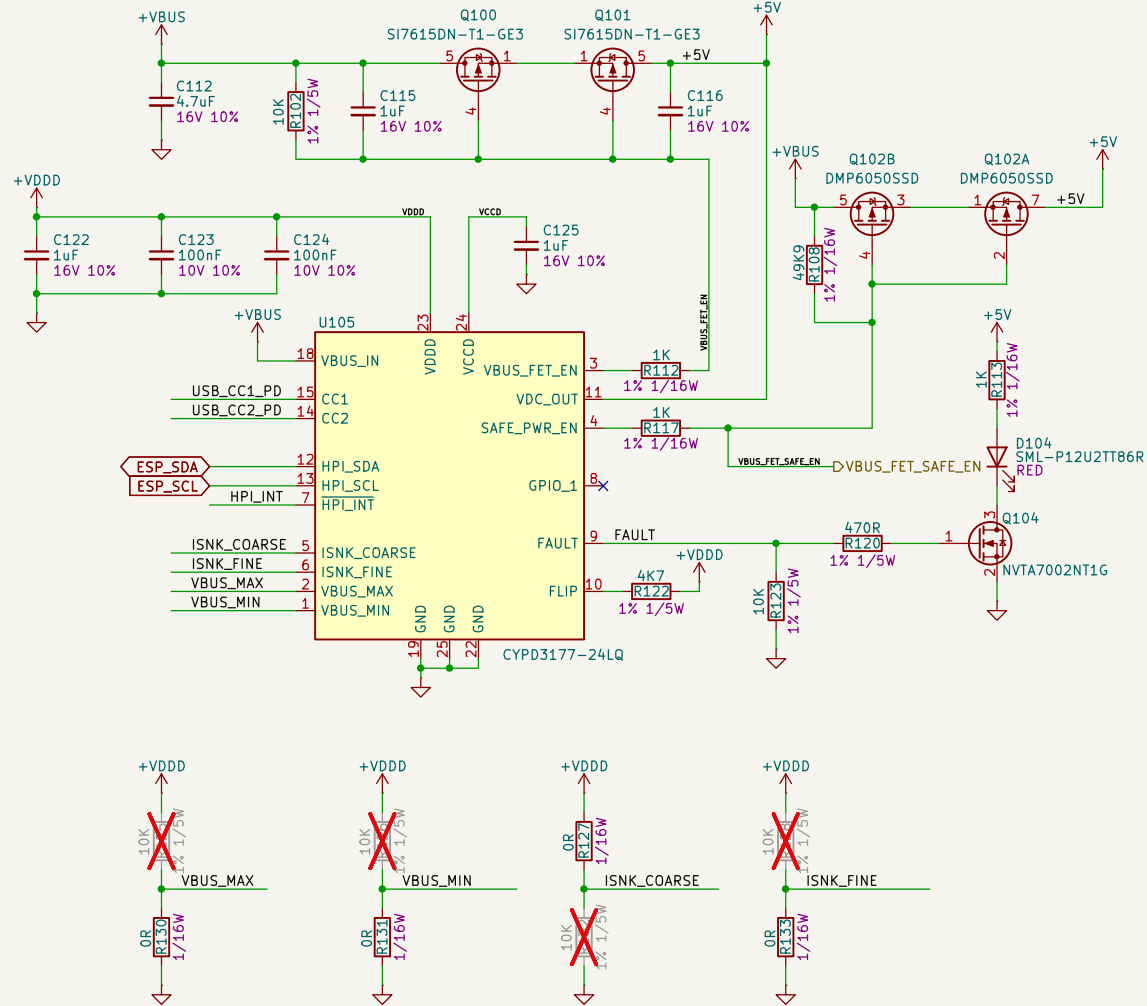
Size: A4 Date: 2025-09-22

KiCad E.D.A. 9.0.0

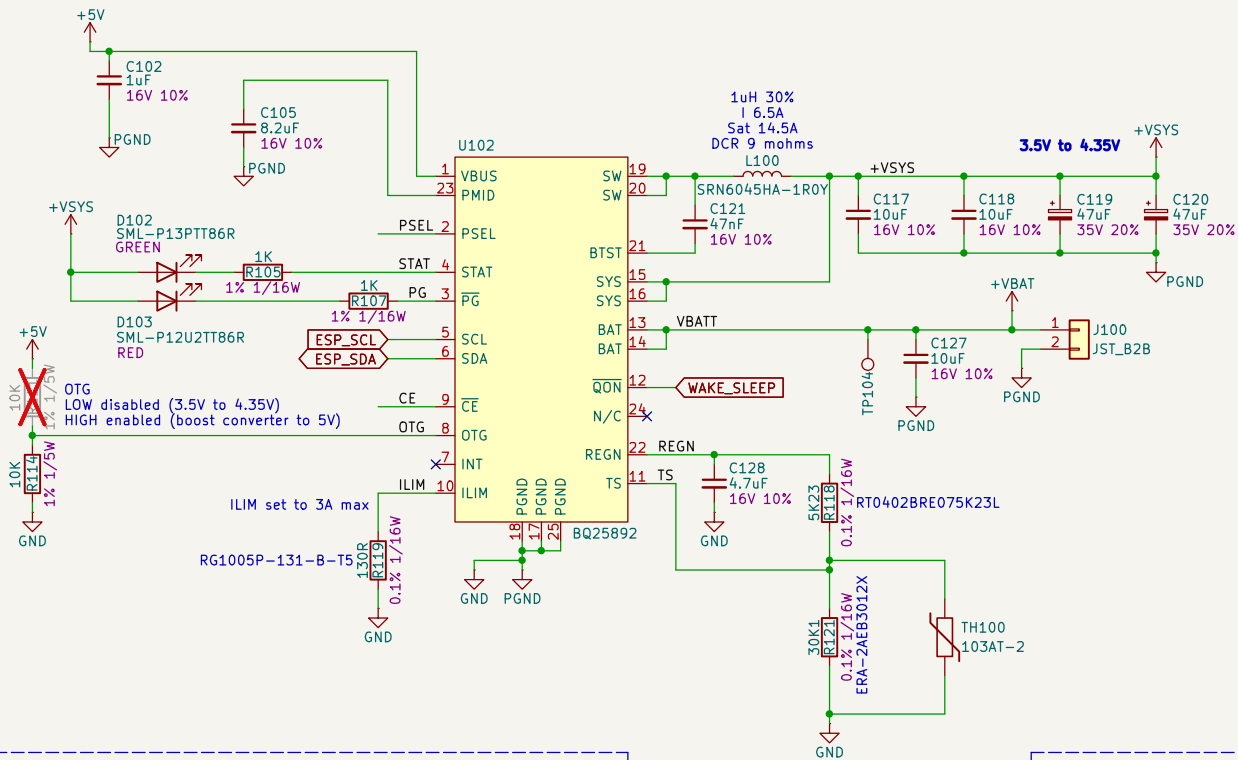
Rev: v0.2

Id: 2/9

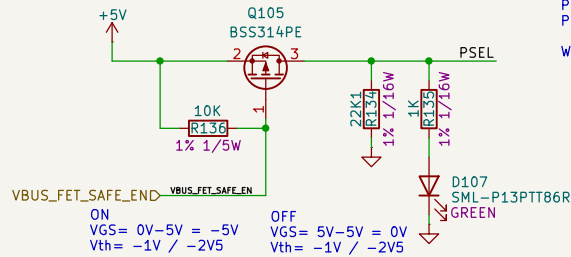
USB C POWER DELIVERY



BATTERY MANAGEMENT IC



HOST OR ADAPTER PSEL



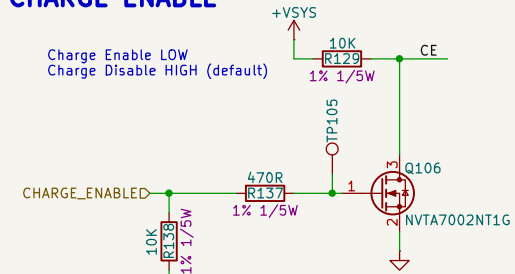
PSEL is controlled by the VBUS_FET_SAFE_EN pin
PSEL HIGH = 500mA
PSEL LOW = 3.25A

When VBUS_FET_SAFE_EN is enabled (Active LOW)
will Pull PMOS gate to 0V and PSEL will see a HIGH sig.

USB <500mA PSEL = HIGH
USB >500mA PSEL = LOW
NO USB default PSEL = LOW

CHARGE ENABLE

Charge Enable LOW
Charge Disable HIGH (default)



Sheet: /BATTERY_MANAGEMENT_IC/
File: BATTERY_MANAGEMENT_IC.kicad_sch

Title: BATTERY_MANAGEMENT_IC

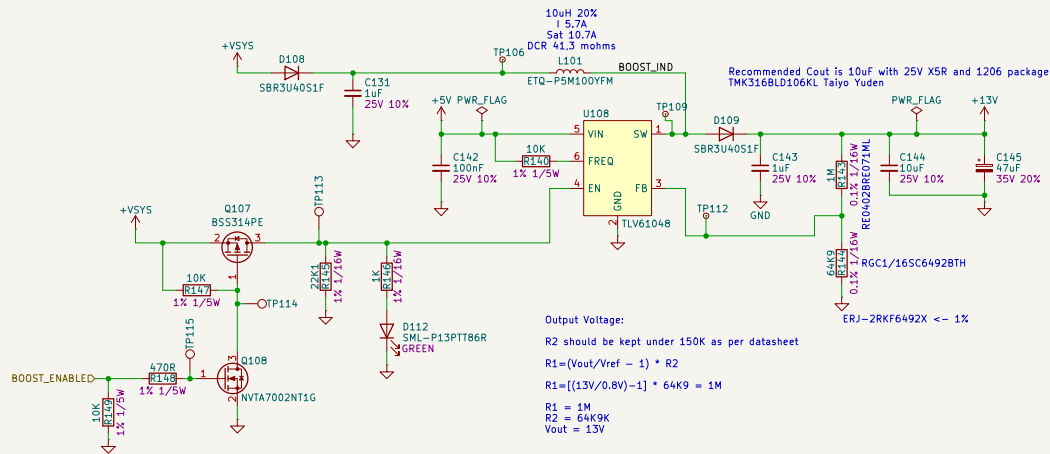
Size: A4	Date: 2025-09-22	Rev: v0.2
KiCad E.D.A. 9.0.0		Id: 4/9

KiCad E.D.A. 9.0.0

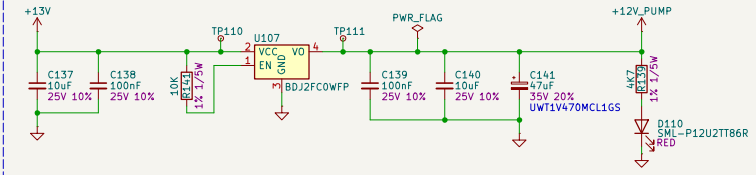
Rev: v0.2

Id: 4/9

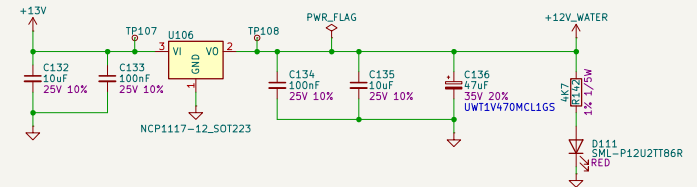
+3V/+5.5V TO +13V BOOST CONVERTER



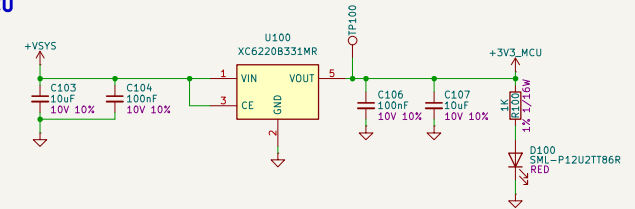
+12V LDO PUMPS



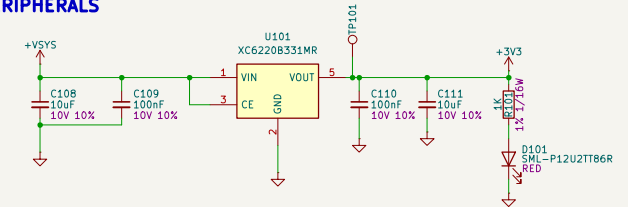
+12V LDO WATER



+3V3 MCU



+3V3 PERIPHERALS



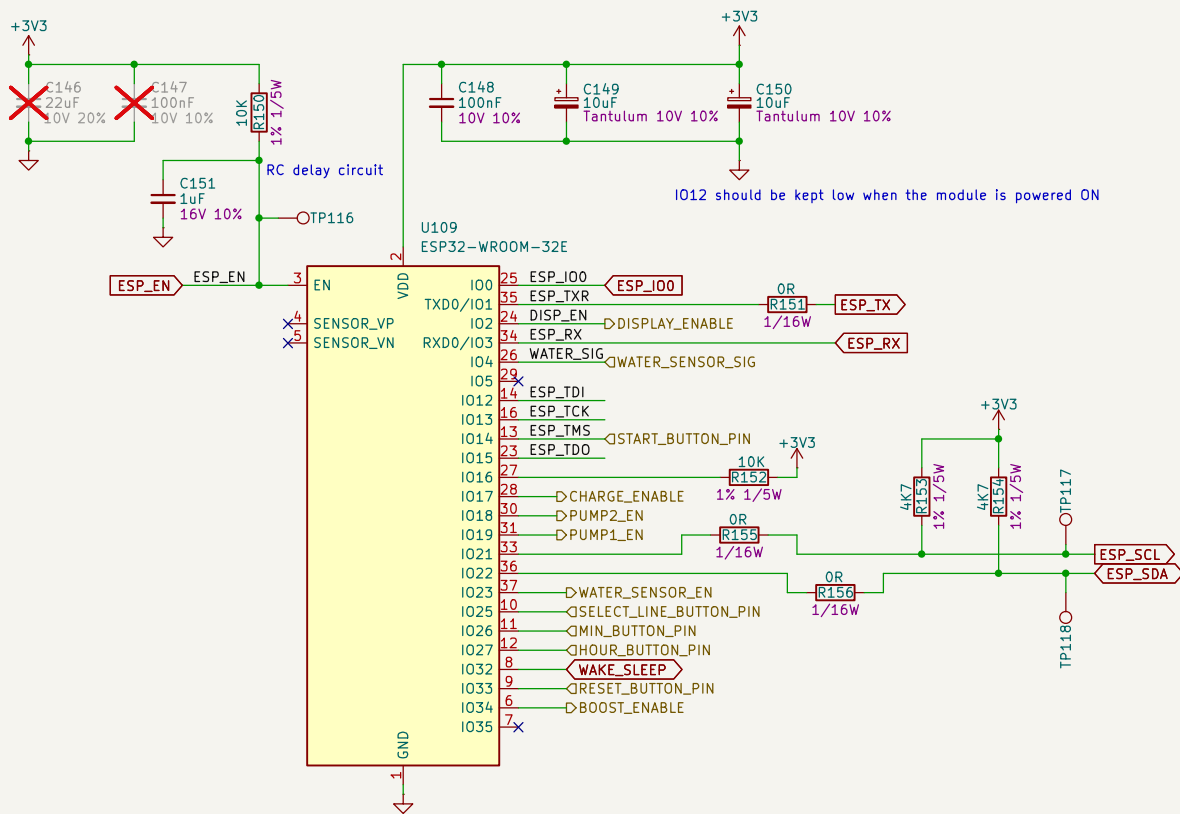
Sheet: /DC_DC_REGULATORS/
File: DC_DC_REGULATORS.kicad_sch

Title: DC_DC_REGULATORS

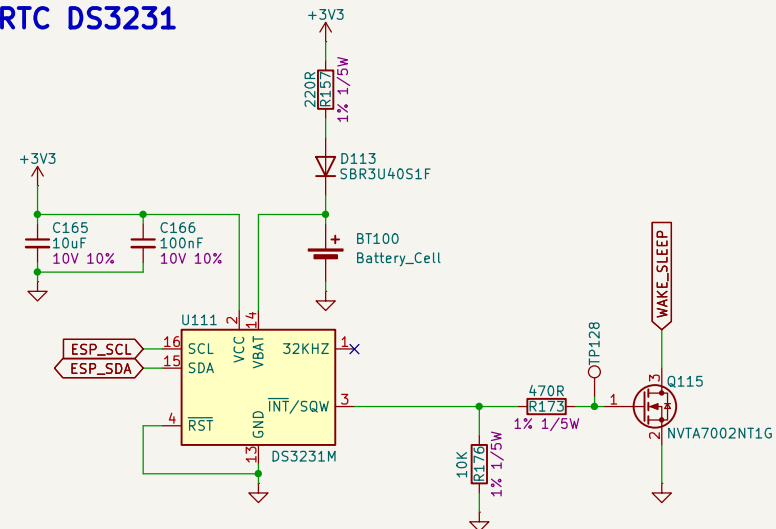
Size: A3 Date: 2025-09-22
KiCad E.D.A. 9.0.0

Rev: v0.2
Id: 5/9

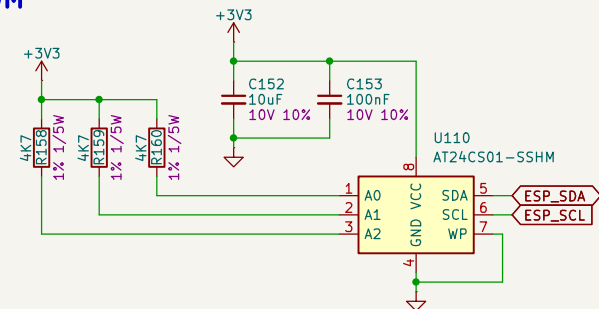
ESP32 WROOM 32E



RTC DS3231



EEPROM



Sheet: /MICROCONTROLLER/
File: MICROCONTROLLER.kicad_sch

Title: MICROCONTROLLER

Size: A4	Date: 2025-09-22
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KiCad E.D.A. 9.0.0

Rev: v0.2

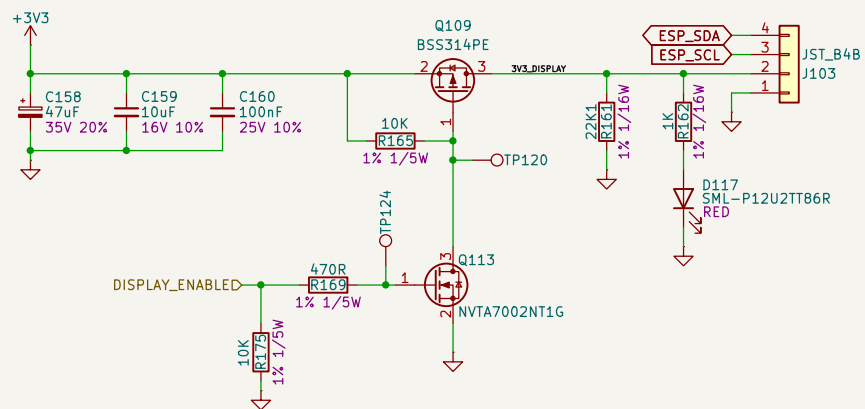
Id: 6/9

PUMP 1

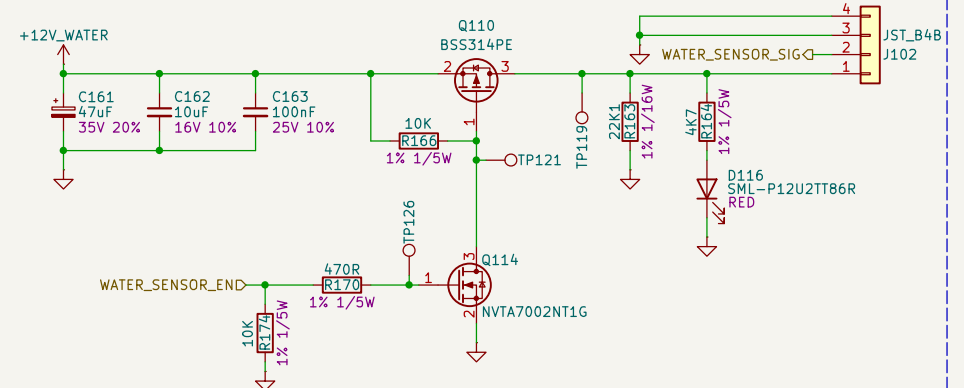
[illegible]

Id: 7/9

OLED DISPLAY ENABLE



WATER SENSOR ENABLE



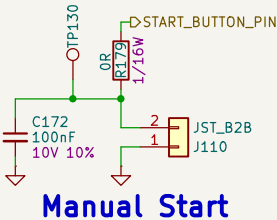
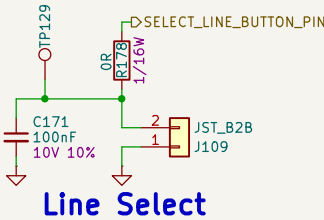
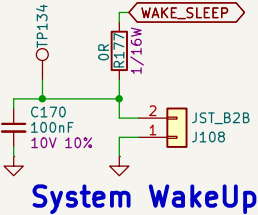
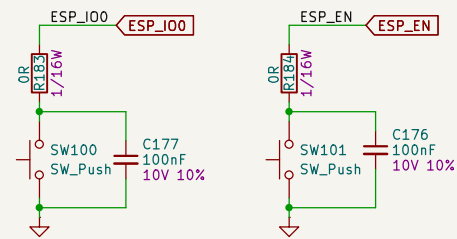
ESP32:
GPIO4 INTERNAL PULL UP TO 3V3
WATER SENSOR:
GPIO4 to PIN2 (THE WATER SENSOR SIGNAL)

Sheet: /DISPLAY_AND_WATER_ENABLE/	
File: DISPLAY_AND_WATER_ENABLE.kicad_sch	
Title: <i>DISPLAY_AND_WATER_ENABLE</i>	
Size: A4	Date: 2025-09-22
KiCad E.D.A. 9.0.0	

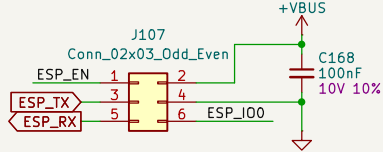
Rev: v0.2
Id: 8/9

USER INTERFACE BUTTONS

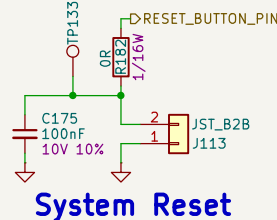
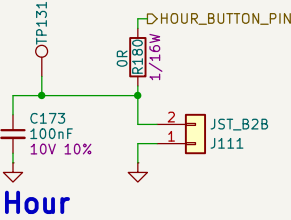
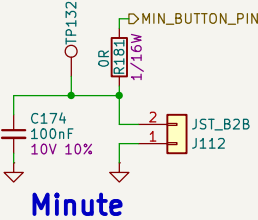
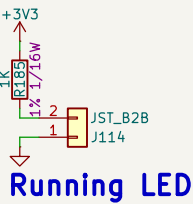
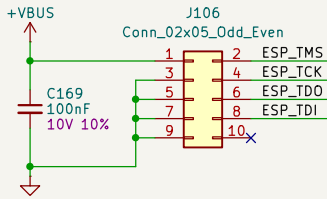
BOOT & RESET



UART_PROG



JTAG



Mounting Holes

